

Engineering Design Report

Fractional Forge — Project a_compact_commercial_airsource_heat_pump

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

£30.00

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

10

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source

[D] Algorithmic estimate or LLM extraction
[E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	FAIL	No regulations listed	0 standards
3. Cost Ceiling Specified	PASS	Target provided	£30.00
4. Spatial Envelope Provided	PASS	Volume established	75 m³
5. Modules Decomposed	PASS	System architecture built	10 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	PASS	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements [A]

1.1 Mission & Use Case

Mission

Not computed

Use Case

Not computed

1.2 Market Context & Timing [B]

Not computed

Not computed

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Not computed	
Product	aroTHERM plus (residential range, up to 15kW) / aroVAIR commercial range
Pricing	Published list pricing for Vaillant commercial heat pump systems is not widely available in the public domain; industry estimates suggest installed costs for commercial systems in the 20–30kW range may fall in the £8,000–£18,000 supply-only range, but this requires verification from current distributor pricing
Tech Specs	Not computed
STRENGTHS Established MCS certification and brand trust with UK installers,R290 refrigerant experience across residential range,Strong UK distribution and training infrastructure,Comprehensive warranty and service network WEAKNESSES Residential R290 range reportedly tops out below 30kW in a single hydronic split unit, potentially requiring cascade or multi-unit configurations,Commercial range may use higher-GWP refrigerants in larger outputs,Premium brand pricing may exceed small commercial budget constraints,Less focused on the specific 25–35kW commercial split segment	
DIFFERENTIATION ANGLE Not computed	

Not computed

Product	Ecodan CAHV / PUHZ commercial hydronic heat pump range
Pricing	Industry sources suggest Mitsubishi Electric Ecodan commercial units in the 28–34kW range carry supply-only prices broadly in the £6,000–£14,000 range depending on configuration; pricing should be verified with current distributor quotes
Tech Specs	Not computed

STRENGTHS

Genuine hydronic split architecture closely matching the proposed product format, MCS certification in place for relevant output ranges, Strong UK installer base and technical support, Proven reliability data across UK commercial installations, Wide operating range and low ambient performance

WEAKNESSES

R32 refrigerant faces increasing regulatory and reputational pressure versus R290, GWP of R32 (approximately 675) is significantly higher than R290 (approximately 3), Supply-only pricing likely significantly above the £4,500 target, Not optimised for sub-35kW small commercial as a distinct market segment

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	Daikin Altherma 3 H HT / Daikin Altherma commercial range
Pricing	Daikin commercial heat pump pricing is not typically published publicly; industry estimates for 25–35kW commercial hydronic units suggest supply-only prices in the £7,000–£15,000+ range, requiring verification
Tech Specs	Not computed

STRENGTHS

Broad product range spanning residential through commercial outputs, High flow temperature capability (up to 65°C) addressing legacy emitter compatibility, Extensive UK distribution and after-sales infrastructure, Strong brand recognition among M&E consultants and specifiers

WEAKNESSES

Primary commercial range uses R32 or R410A, not R290, R290 product development appears focused on residential outputs below 20kW, Pricing likely well above the £4,500 target at this output, Large organisation may respond slowly to niche small commercial segment requirements

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	Aquarea T-CAP / Aquarea High Performance commercial range
Pricing	Industry sources suggest Panasonic Aquarea commercial units in the 30kW class carry supply-only prices broadly in the £5,500–£12,000 range; verification against current distributor pricing is required
Tech Specs	Not computed

STRENGTHS

One of the closest existing competitors in output range and architecture to the proposed product, Hydronic split format well established in UK commercial market, Competitive pricing within the commercial heat pump sector, MCS certification for applicable products, Good low ambient temperature performance

WEAKNESSES

R32 refrigerant carries regulatory risk versus R290, Brand recognition in UK small commercial HVAC sector reportedly lower than Mitsubishi or Daikin, Installer training and support network less extensive than tier-one competitors, Limited differentiation on refrigerant sustainability credentials

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	OCHSNER Europa Mini / GMWW range (commercial hydronic heat pumps)
Pricing	Ochsner commercial heat pump pricing is not widely published in the UK market; the brand operates primarily through specification and project channels; pricing should be sought directly from UK distributors
Tech Specs	Not computed

STRENGTHS

European manufacturer with experience in commercial hydronic heat pump engineering, Some product lines developed with low-GWP refrigerants, Specification-led sales approach suits M&E consultant engagement, Engineering heritage in commercial heat pump applications

WEAKNESSES

Limited UK market presence and installer familiarity, MCS certification status in UK requires verification, Distribution and service infrastructure in UK less developed than Japanese OEM competitors, Brand not widely recognised by UK small commercial installers

DIFFERENTIATION ANGLE

Not computed

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets ^[A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	500
Target Cost Ceiling	£30.00
Max Mass	Not computed

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2. Regulatory & Compliance

No regulatory standards identified for this brief.

3. Sizing & Spatial Optimisation ^[D]

INFEASIBLE SPATIAL ALLOCATION

Modules exceed available envelope.

System Envelope

Internal Dimensions (W×D×H)	5000 × 5000 × 3000 mm
Internal Floor Area	25 m ²
Internal Volume	75 m ³
Rules Domain	hardware_product

Module Allocation Zone Table

Module	L × W × H (mm)	Area (m ²)	Mount
outdoor_chassis_acoustic_enclosure_module	2236 × 2236 × 2400	5	floor
air_to_refrigerant_evaporator_coil_module	2236 × 2236 × 2400	5	floor
active_airflow_management_module	2236 × 2236 × 2400	5	floor
hermetic_r290_compressor_sub_assembly	2236 × 2236 × 2400	5	floor
refrigerant_expansion_control_module	2236 × 2236 × 2400	5	floor
refrigerant_to_water_brazed_plate_heat_exchanger_bphe_	2236 × 2236 × 2400	5	floor
outdoor_atex_electrical_inverter_drive_module	2236 × 2236 × 2400	5	floor
indoor_hydraulic_circulation_module	2236 × 2236 × 2400	5	floor

indoor_auxiliary_heat_expansion_module	2236 × 2236 × 2400	5	floor
master_control_bms_gateway_module	2236 × 2236 × 2400	5	floor

Conflicts

- Total module footprint (50.0m²) exceeds envelope budget (21.3m²)

Data Sources

[Deterministic Solver] 3D Box Packing Constraints Engine

Overall Section Grade: B

4.1 Outdoor Chassis & Acoustic Enclosure Module ^[D]

Maturity: draft

Purpose

Provides structural integrity, weather protection, and acoustic dampening for the highly sensitive outdoor refrigeration components.

Why It Matters

Not computed

Technical Description

The outer chassis is fabricated from folded galvanised steel finished with a polyester powder coat of minimum 60 µm dry film thickness . This coating system is rated for C4 and C5 corrosion categories per ISO 12944, covering coastal and heavy industrial environments. The architecture physically segregates the compressor bay within an ATEX-compliant enclosure to isolate R290 (propane) from electrical ignition sources.

All internal faces of the compressor housing are lined with flame-retardant EPDM elastomeric foam panels. These panels attenuate broadband mechanical noise to achieve a target sound power level of 65–70 dB(A) at 1 m . The baseplate carries elastomeric anti-vibration mounts rated for the compressor's operating frequency range of 25–80 Hz , decoupling rotational impulse from the steel frame.

The lower casing incorporates fixed, non-closing louvred vents positioned within 100 mm of the base pan . R290 vapour has a relative density of approximately 1.55 relative to air and accumulates at floor level. These vents provide passive dilution ventilation to disperse any leaked refrigerant per BS EN 378-3 requirements, preventing accumulation above 10% of the Lower Flammability Limit (LFL) of 21,000 ppm .

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.2 Air-to-Refrigerant Evaporator Coil Module [D]

Maturity: draft

Purpose

Absorbs thermal energy from the ambient outdoor air to boil the liquid R290 into a gas.

Why It Matters

Not computed

Technical Description

The forced-draft air-to-refrigerant heat exchanger is sized for a continuous thermal absorption duty of 30 kW at a design ambient condition of 0°C dry bulb . The coil matrix comprises internally grooved copper tubes, typically 9.52 mm outer diameter, mechanically expanded into aluminium fins at a pitch of 2.2 mm . This fin spacing deliberately delays frost bridge formation during the critical ambient band of -5°C to +5°C, where frosting rate is highest .

The aluminium fin surface receives a hydrophilic epoxy coating approximately 3–5 µm thick . This reduces water droplet contact angle below 30°, promoting rapid condensate sheet-drainage and preventing water retention that would otherwise accelerate fin corrosion in the UK's high-humidity climate. A multi-port copper distributor header feeds R290 uniformly across all parallel circuits, maintaining mass flow deviation below ±5% between ports .

The coil assembly sits on thermally isolating glass-fibre reinforced nylon brackets to interrupt conducted heat loss to the steel chassis. A 230 V, 150–250 W resistive heater cable element runs through the stainless steel drip tray below the coil , activating during each defrost cycle to prevent ice accretion blocking condensate drainage.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.3 Active Airflow Management Module [D]

Maturity: draft

Purpose

Drives large volumes of outdoor air across the evaporator coil to maintain a high temperature differential and prevent frost stalling.

Why It Matters

Not computed

Technical Description

The module comprises two axial fans, each 600–800 mm in diameter, driven by IP54-rated Electronically Commutated (EC) motors sourced from manufacturers such as ebm-papst or Ziehl-Abegg . Each motor delivers a typical shaft power of 0.5–1.5 kW depending on duty point . Variable-speed control via 0–10 V or PWM signal adjusts fan speed continuously against ambient temperature and compressor load, reducing parasitic electrical consumption by up to 30% compared to fixed-speed alternatives .

Fan blades are moulded from glass-fibre reinforced composite or cast aluminium alloy and feature swept-blade aerofoil geometry. This geometry reduces trailing-edge vortex shedding noise by approximately 3–5 dB(A) compared to flat-blade equivalents at the same air volume flow . Each blade set is dynamically balanced to an ISO 1940 Grade G2.5 tolerance .

The fans are housed within aerodynamically profiled shrouds moulded from impact-resistant Acrylonitrile Butadiene Styrene (ABS), which reduces inlet flow separation losses and maintains a stable pressure rise across the 30 kW coil face. Powder-coated mild steel bird guard grilles with a maximum 25 mm aperture protect the composite blades from foreign object contact .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.4 Hermetic R290 Compressor Sub-Assembly [D]

Maturity: draft

Purpose

Mechanically compresses the low-pressure R290 gas into a high-pressure, high-temperature gas to drive thermal energy into the hydronic circuit.

Why It Matters

Not computed

Technical Description

The compression element is a DC inverter-driven scroll compressor, such as the Copeland YHV or GMCC commercial series, chemically qualified for R290 and rated for continuous operation across a condensing pressure range of approximately 18–26 bar(g) . The inverter modulates compressor frequency between approximately 30 Hz and 100 Hz, varying capacity from 30% to 100% of nominal 30 kW duty to match the building's dynamic thermal demand .

The compressor's steel shell is enclosed in a 50 mm thick wrap-around jacket moulded from flame-retardant melamine foam , reducing radiated casing noise by approximately 8–12 dB(A) at the shell surface . The compressor rests on elastomeric vibration isolators selected and pre-loaded for the nominal 40 kg compressor mass . These isolators are tuned to a natural frequency below 5 Hz to provide greater than 90% vibration isolation efficiency across the operating speed range .

A 50–150 W ATEX-rated crankcase heater element is wound around the compressor sump . During shutdown periods in ambient temperatures below +10°C, this heater maintains sump oil temperature above the R290 saturation point, preventing liquid refrigerant migration into the POE lubricating oil. A flooded oil charge at start-up would drive cylinder pressure spikes sufficient to fracture scroll tip seals within seconds .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.5 Refrigerant Expansion & Control Module [D]

Maturity: draft

Purpose

Controls the phase state, operating pressure, and flow direction of the R290 refrigerant throughout the thermodynamic cycle.

Why It Matters

Not computed

Technical Description

The primary flow control device is an Electronic Expansion Valve (EEV) driven by a bipolar stepper motor, sourced from manufacturers such as Danfoss or Carel. The stepper motor adjusts the internal orifice in discrete steps, typically 480 or 960 steps across the full operating range , to maintain suction superheat within a target band of 5–8 K entering the compressor . This prevents liquid carry-over to the compressor while maximising the evaporator's active refrigerant two-phase length.

A pilot-operated 4-way reversing valve, constructed from brass with polytetrafluoroethylene (PTFE) internal seals, physically redirects hot discharge gas from the compressor into the outdoor coil during timed hot-gas defrost cycles. Defrost initiation is based on a combination of evaporator air-side differential pressure and elapsed run-time logic .

All pipe joints within this module are silver-alloy brazed under a flowing dry nitrogen purge at a minimum pressure of 0.5 bar(g) to prevent copper oxidation inside the circuit . Flare fittings are excluded entirely. Solid-core hermetic filter driers, rated for R290 service, are sized to hold a minimum of 3% by weight of the total refrigerant charge in desiccant . High and low-side pressure transducers with a 4–20 mA output are mounted on machined Schrader-ported bosses with integral shut-off cores.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.6 Refrigerant-to-Water Brazed Plate Heat Exchanger (BPHE) ^[D]

Maturity: draft

Purpose

Transfers thermal energy from the high-temperature R290 discharge gas into the building's circulating water or glycol hydronic loop.

Why It Matters

Not computed

Technical Description

Heat transfer between the refrigerant and hydronic circuits is achieved in a compact Brazed Plate Heat Exchanger (BPHE) constructed from 316-grade stainless steel plates vacuum-brazed with pure copper at approximately 1,083°C . The plate count and corrugation geometry are selected to achieve a mean temperature approach of 3–5 K between refrigerant condensing temperature and hydronic water outlet temperature at 30 kW duty . Manufacturers such as SWEP or Alfa Laval supply units to this standard.

The alternating counter-flow arrangement passes R290 gas and hydronic water through chevron-corrugated micro-channels, generating turbulence at Reynolds numbers typically below 500 that nonetheless achieve overall heat transfer coefficients of 3,000–5,000 W/m²K . The complete BPHE assembly is encased in 19–25 mm thick closed-cell elastomeric foam insulation, such as Armaflex Class O , limiting heat loss to the outdoor air to below 50 W at 0°C ambient .

An inline flow switch, either paddle-actuated or ultrasonic type, is installed within 300 mm of the hydronic water outlet . This device is calibrated to a minimum flow threshold of approximately 60% of design flow rate . A drop below this threshold triggers an immediate compressor lockout to prevent freezing of the water-side channels.

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.7 Outdoor ATEX Electrical & Inverter Drive Module ^[D]

Maturity: draft

Purpose

Powers the inverter compressor and manages high-voltage distribution while eliminating electrical ignition risks within the outdoor refrigerant hazard zone.

Why It Matters

Not computed

Technical Description

The electrical enclosure is a galvanised steel box rated to IP54 minimum, mounted in the upper chassis section at least 300 mm above the base pan to remain outside the primary R290 gas accumulation zone. Internal components are spatially separated from the lower ATEX-compliant compressor bay by a sealed bulkhead. EMC line filters and series chokes are fitted on all incoming power phases to limit conducted emissions to Class C2 per EN 61000-3-2 .

The high-power 3-phase inverter board comprises IGBT modules rated for the full 30 kW compressor load with a minimum 20% derating margin applied . AC-to-DC rectifier bridges and DC link electrolytic capacitors are rated at 450 V DC and specified for a design life of 60,000 hours at 40°C . A solid-state soft-start algorithm limits inrush current to no more than 150% of full-load current during grid energisation .

All contactors within the enclosure are either hermetically sealed solid-state devices or explicitly ATEX-certified to Group IIA, Temperature Class T2 for propane service . A hardwired safety relay monitors a factory-installed catalytic bead gas sensor. When the sensor detects propane concentrations reaching 10% of the LFL (2,100 ppm) , the relay immediately de-energises all non-ATEX ignition sources and activates a dedicated chassis purge fan.

Bill of Materials (BOM) ^[D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.8 Indoor Hydronic Circulation Module [D]

Maturity: draft

Purpose

Pumps heated fluid from the outdoor heat exchanger into the building's thermal buffer tank and distribution system.

Why It Matters

Not computed

Technical Description

The circulation module is housed within the wall-mounted indoor hydrobox and is driven by a variable-speed Electronically Commutated Motor (ECM) circulator pump, such as the Grundfos MAGNA3 or Wilo Stratos MAXO . Pump speed is modulated via a 0–10 V or PWM control signal to maintain a constant delta-T of 5–7°C across the outdoor BPHE at all operating points . At design flow rate for a 30 kW system, the circulator typically delivers 0.86–1.44 l/s at a developed head of 2–4 m .

A high-intensity magnetic dirt separator is installed as a mandatory in-line component immediately upstream of the outdoor BPHE water inlet connection . The separator's permanent magnet assembly captures ferrous magnetite particles down to 6 µm, protecting the BPHE channels from progressive fouling . Heavy-duty full-bore brass isolation ball valves are fitted on the inlet and outlet of the separator and the pump to permit servicing without draining the entire system volume.

Automatic Air Vent (AAV) valves, constructed from DZR brass, are fitted at all high points of the hydrobox pipework . These valves continuously vent dissolved micro-bubbles with an automatic float-controlled seal, maintaining dissolved gas content below levels that trigger pump cavitation, which begins when local pressure drops below the fluid vapour pressure at the impeller eye .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.9 Indoor Auxiliary Heat & Expansion Module [D]

Maturity: draft

Purpose

Safely absorbs hydronic fluid thermal expansion and provides automated bivalent backup heating at extreme ambient conditions.

Why It Matters

Not computed

Technical Description

A hydro-pneumatic diaphragm expansion vessel, typically sized between 50 and 100 litres for a 30 kW system with an estimated circuit volume of 80–150 litres , is connected to the hydronic circuit via a dedicated non-return-valved connection. The vessel contains a butyl rubber diaphragm backed by a dry nitrogen pre-charge, typically set at 1.0–1.5 bar(g) . The diaphragm accommodates the volumetric expansion of the water-glycol mixture from a cold-fill temperature of 10°C to a maximum operating temperature of 65°C without exceeding the 3 bar(g) system design pressure limit .

An inline electric immersion heater manifold, constructed from Incoloy 825 alloy elements for corrosion resistance in glycol solutions , provides bivalent supplementary heat in stepped increments of 3 kW up to a total of 9 kW . The controller activates these elements automatically when outdoor ambient temperature drops below 7°C and the heat pump alone cannot maintain the target flow temperature .

A 3 bar(g) brass pressure relief valve (PRV) is hard-wired into the pipework immediately adjacent to the expansion vessel and discharges via a copper tundish to a visible safe location , complying with Part G of the Building Regulations. A sealed Bourdon-tube mechanical pressure gauge, readable without electrical power, is mounted at the same manifold point to allow manual pressure verification during commissioning and maintenance.

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

4.10 Master Control & BMS Gateway Module [D]

Maturity: draft

Purpose

Orchestrates the thermodynamic cycle, calculates weather-compensated flow temperature targets, and provides data exchange with the building management system.

Why It Matters

Not computed

Technical Description

The primary sequence controller is a ruggedised commercial HVAC microprocessor platform, such as the Carel pCO series or Siemens Climatix , operating on a 24 V DC supply and rated for an ambient operating range of 20°C to +60°C . The weather compensation algorithm samples the outdoor NTC thermistor every 30 seconds and calculates a target flow temperature on a defined heating curve. A typical curve reduces flow temperature from 55°C at 10°C ambient to 35°C at +15°C ambient, maximising COP throughout the season .

The communications module supports RS485 physical layer at baud rates up to 115,200 bps , carrying Modbus RTU, Modbus TCP/IP, and BACnet MS/TP or BACnet/IP protocols. This multi-protocol capability satisfies UK commercial M&E specification requirements set by CIBSE Guide H . The gateway maps a minimum of 50 data objects, including real-time COP, compressor frequency, flow and return temperatures, and fault codes, for interrogation by the building-wide BMS .

The wall-mounted HMI panel features a high-contrast colour TFT LCD touchscreen of at least 7 inches diagonal , displaying live system pressures, temperatures, and COP values without requiring a laptop or proprietary service tool. A Class 1 Pt1000 resistance temperature sensor wiring harness connects the controller to a minimum of 8 sensor points across both indoor and outdoor units .

Bill of Materials (BOM) [D]

Part #	Description	Supplier	Grade	Cost
No parts defined.				

Data Sources

[LLM Architecture] Decomposition Logic

[LLM BOM] Parts expansion

Overall Section Grade: D

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
Outdoor Chassis & Acoustic Enclosure Module	Structural shearing during extreme wind loads.	Not computed	Not computed	4×2=8	Integrated cross-braced structural skeleton and heavy-gauge baseplate geometry.
Outdoor Chassis & Acoustic Enclosure Module	Accumulation of highly flammable R290 gas.	Not computed	Not computed	5×2=10	Natural drafting base design with absolutely no gas traps or enclosed low points.
Air-to-Refrigerant Evaporator Coil Module	Total loss of R290 charge via ruptured coil u-bend.	Not computed	Not computed	4×2=8	Helium/X-ray leak testing of all factory brazed joints and robust vibration damping.
Air-to-Refrigerant Evaporator Coil Module	Coil crush damage during transit or rigging.	Not computed	Not computed	3×3=9	Heavy-duty cardboard and timber crate design with internal structural bracing around coil.
Active Airflow Management Module	Catastrophic blade disintegration discharging shrapnel.	Not computed	Not computed	4×1=4	Vigorous dynamic balancing of fan assemblies and robust high-tensile steel bird guards.
Active Airflow Management Module	High acoustic emissions from turbulent stall.	Not computed	Not computed	3×3=9	Deep acoustic shrouds, swept blade design, and vibration-isolated motor brackets.
Hermetic R290 Compressor Sub-Assembly	Ignition of leaked R290 gas by an internal compressor electrical arc.	Not computed	Not computed	5×1=5	100% sealed, ATEX-certified electrical terminal box on the compressor body.

Hermetic R290 Compressor Sub-Assembly	Compressor overheating under extended intense thermal lift payload.	Not computed	Not computed	3×2=6	Liquid injection cooling ports into the scroll casing and strict software envelope constraints.
Refrigerant Expansion & Control Module	High-pressure R290 leakage from a failed pipe joint.	Not computed	Not computed	4×2=8	Strict robotic brazing where possible, elimination of mechanical flare connections, factory helium leak testing.
Refrigerant Expansion & Control Module	Catastrophic system over-pressure event (e.g. blocked condenser).	Not computed	Not computed	4×1=4	Hard-wired mechanical high-pressure safety cut-out switches totally independent of software control.
Refrigerant-to-Water Brazed Plate Heat Exchanger (BPHE)	Flammable R290 migration into an occupied building via the water loop.	Not computed	Not computed	5×1=5	Use of asymmetric/double-wall capable BPHE rated over 45 bar tested burst pressure, external AAVs.
Refrigerant-to-Water Brazed Plate Heat Exchanger (BPHE)	BPHE freezing, cracking plates and destroying the entire closed loop system.	Not computed	Not computed	4×2=8	Redundant flow sensor interlocks; mandatory requirement for 25% propylene glycol dosing in the water circuit.
Outdoor ATEX Electrical & Inverter Drive Module	Exposed electrical arcing igniting a pooling propane leak.	Not computed	Not computed	5×1=5	Absolute spatial segregation of the electrical box, solid-state switching, and active ATEX purge fan.
Outdoor ATEX Electrical & Inverter Drive Module	M&E Operator electrocution during maintenance of active 400V capacitors.	Not computed	Not computed	5×1=5	Integrated, lockable isolator switch, dead-front polycarbonate panel design, active discharge bleed resistors.
Indoor Hydronic Circulation Module	Major pressurized water leak causing significant indoor commercial property damage.	Not computed	Not computed	3×2=6	High-quality mechanical compression seals, factory hydro-testing, integrated hydrobox drip tray with float alarm.
Indoor Hydronic Circulation Module	Silent pump stall leading to near-instant freezing in the outdoor unit BPHE.	Not computed	Not computed	4×2=8	Pulse-width modulation feedback wire immediately alerts Master Controller to ECM fan failure.

Indoor Auxiliary Heat & Expansion Module	Discharge of 65°C scalding pressurized water from PRV into occupied utility spaces.	Not computed	Not computed	3×2=6	Mandatory visual tundish piped securely in rigid copper to low-level commercial drainage points.
Indoor Auxiliary Heat & Expansion Module	Auxiliary heater dry-fire causing critical component melting and local fire risk.	Not computed	Not computed	4×1=4	Hardwired, manual-reset mechanical thermal high-limit stat and software flow switch interlock.
Master Control & BMS Gateway Module	Network cybersecurity breach via exposed or default-password BACnet IP commercial port.	Not computed	Not computed	4×2=8	Secure read-only baseline configurations, strict gateway passwords, and advising IT on hardware VLAN segregation.
Master Control & BMS Gateway Module	Total logic control failure leading to unrecognized system freezing off-hours.	Not computed	Not computed	4×1=4	Hardwired, mechanically actuated freeze protection pipe thermostat completely independent of software.

Data Sources

[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	10 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success